

Research Question

How do quarterback air-oriented passing styles (deep/air yards and aggressiveness) relate to drive success and turnover risk in the 2023 NFL season?

Evidence Plan

Data: Use `nflfastR` 2023 play-by-play (`nflfastR_pbp_2023_clean.csv`) and the QB weekly/season exports (`qb_weekly_2023.csv`, `qb_season_2023.csv`).

Key variables:

- Passing style: average air yards per attempt, air-yard share, aggressiveness proxy (deep pass rate), CPOE, EPA/dropback.
- Outcomes: drive success (points per drive, TD rate), turnover risk (INT rate, interceptions per attempt).
- Controls: down/distance mix, score differential (situational aggressiveness), opponent defense.

Methods:

- Descriptive: correlate air yards and deep rate with INT rate and EPA/dropback across QBs; plot quartiles of air yards vs INT rate.
- Regression: QB-level (or mixed-effects on play/drive level) modeling INT rate and EPA/dropback as functions of air yards per attempt and CPOE, controlling for down/distance and score differential.
- Robustness: repeat with drive-level outcomes (points per drive) and exclude garbage time (score differential within 14 points).

Evidence sources: Primary: stats from `nflfastR` files above. Secondary: 1–2 academic/analytics articles on air yards/aggressiveness vs efficiency (e.g., QB air yards and EPA, aggressiveness vs turnover risk).